Journal of Public Health (2024) 32:2091–2100
https://doi.org/10.1007/s10389-023-01964-8

ORIGINAL ARTICLE

Impact of smartphone use at bedtime on sleep quality and academic activities among medical students at Al -Azhar University at Cairo

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Received: 29 March 2023 / Accepted: 26 May 2023 / Published online: 15 June 2023
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Abstract

Aim Smartphone use has become prevalent and rapidly expanding. Therefore, it is essential to determine the potential adverse effects. This study aims to assess the prevalence of smartphone use at bedtime and its association with sleep quality and educational-related activities among medical students.

Subject and methods The current cross-sectional study included 1184 male and female undergraduate medical students at Al-Azhar University in Cairo. We used a self-administered questionnaire to collect data on personal characteristics, smartphone use, and academic activities. The Pittsburgh sleep quality index ((PSQI) was used for the assessment of sleep quality.

Results The result revealed that bedtime smartphone use was highly prevalent (96.5 %), with nearly half (46.9%) of studied medical students using them for more than 2 hours. Social media was at the top of the usage list (74.2%), followed by communication (48.8%). Smartphone users at bedtime had a significantly higher prevalence of poor sleep quality, increased sleep latency, difficulty in maintaining sleep, and shorter sleep duration compared to non-user (p-value<0.05). The probability of poor sleep quality is significantly increased by bedtime smartphone use, regular coffee drinking, hostel residence, and proximity of the phone during sleep, whereas the likelihood of poor sleep quality is significantly decreased by physical activity. A significant association between poor sleep quality and academic performance was also reported.

Conclusion This study emphasizes the negative impact of smartphone use at bedtime on sleep quality and quantity, which in turn has a negative effect on academic performance. Therefore, medical students are strongly encouraged to limit their smartphone use before bedtime for better sleep and academic performance.

Keywords Smartphone · Sleep quality · Academic activities · Medical students · Pittsburgh sleep quality index (PSQI)

Background

Smartphones have transformed global lifestyles and become an integral part of many people's lives, particularly adolescents (Okela 2023), due to their connectivity and information access, which enable web browsing, entertainment, and communication for users. Moreover, smartphones are increasingly becoming an essential educational tool for all students (Hashmi et al. 2019). Egypt has 75.66 million cellular connections as of January 2022 (Kemp 2022), and 70% of internet users are university students (MCIT 2021).

However, the constant use of and addiction to cell phones can have negative physical, psychological, and social effects on individuals (Naveenta et al. 2016; Okasha et al. 2021). In Egypt, Eldesokey (2021) found that smartphone addiction was prevalent among medical students at a rate of 53.6%, and Mohamed and Moustafa (2021) reported a significantly higher rate of 74.7%.

As a key indicator of problematic issues, smartphone use before bedtime was prevalent among users (Paik et al. 2019). Increased use of smartphones before bedtime directly predisposes users to sleep disturbances and psychological distress (Alshobaili and AlYousefi 2019). Prolonged use of mobile phones at night has a significant impact on circadian rhythms. The light emitted from smartphone screens is rich

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in the blue portion of the visible light spectrum and has a potent stimulating effect on retinal ganglion cells containing melanopsin. Stimulation of these cells suppresses melatonin secretion and reduces sleep quality (Randjelović et al. 2019; Rathakrishnan et al. 2021).

Sleep is necessary for all mental, physical, and emotional daily activities. Chronic sleep deprivation impairs cognitive functions such as memory, concentration, and response time. Among medical students, good sleep quality is vital for academic development due to the nature of their study and the vast number of tasks they must complete ((Dewald et al. 2010; Al Shammari et al. 2020; Rathakrishnan et al. 2021). Poor sleepers experience more daytime difficulties as regards fatigue, sleepiness, poor cognition, and attention during class than those who sleep well (Alapin et al. 2000; Orzech et al. 2011).

The aforementioned associations can be explained by the stressor-strain-consequence (SSO) theory. This theory assumes that a stressor indirectly impacts the outcome, and the strain usually acts as an intermediary between the stressor and the outcome (Gary and Koeske 1993). Therefore, late-night use of smartphones (as a stressor) negatively affects academic performance (as an outcome) through the intervening strain effect of worse sleep quality and cognitive function depletion (Luqman et al. 2020).

Numerous studies demonstrate that the majority of mobile users suffer from sleep deprivation and increased stress, impairing their cognitive and learning abilities (Naveenta et al. 2016; Rupani et al. 2016; Bouazza et al. 2023). Until now, few studies have been conducted on medical students using smartphones before bedtime and their impact on sleep disturbances and academic performance. Hence, the current study aimed to assess the prevalence of smartphone use at bedtime and its association with sleep quality and related educational activities among undergraduate medical students in the Faculty of Medicine, Al-Azhar University, Cairo.

Subjects and methods

An observational cross-sectional study was performed from March 2021 to March 2022, including 1184 male and female undergraduate medical students from all academic grades of both faculties of medicine (boys and girls), Al-Azhar University at Cairo.

Sample size and sampling technique

The sample size was calculated according to the following formula (n=[(Zα/2)2* P(1- P)]/d2), assuming a proportion of medical students with poor sleep quality of 55% (Jahrami et al. 2020), a confidence level of 95%, and an absolute precision of 0.03. Accordingly, the minimum required sample size was 1056. Following the addition of 15% to account for attrition, the final sample size was 1184. The sample size was proportionately allocated to both faculties of boys and girls. The studied participants were selected based on a stratified cluster sampling technique; first, students were stratified according to academic grades, and then 1 or 2 classes (clusters) were chosen randomly from each academic grade. Except for subjects diagnosed with sleep disorders, chronic respiratory illnesses, and physical or mental illnesses that affected sleep, all students in the selected clusters were eligible to participate in the study.

Study tools

A Self-administered questionnaire consisting of 4 parts was used: The first section consisted of questions regarding demographic information (i.e., age, sex, academic year, living status, and body mass index) and lifestyle behaviors (i.e., consumption of caffeinated drinks after 4 p.m., smoking, and exercise for half an hour at least three times/week).

The 2nd part assessed smartphone use behaviors in which students were asked about the average time of smartphone use/day, smartphones use within two hours before bedtime, and the duration of their usage at bedtime (<1 h, 1–2 h, more than 2 h). In addition, they were asked about the most common reason for using their cellphones before bed, the adjustment of the screen's brightness, whether they kept their smartphones under the pillow or by the bed during sleep, and whether they switched off their phones during sleep.

In the 3rd part, the PSQI was used to assess the quality of sleep (Buysse et al. 1989). It consists of 19 items divided into 7 components, including questions about sleep duration, sleep disturbance, sleep latency, sleep efficiency, use of sleep medication, daytime dysfunction, and subjective sleep quality. Each component was scored from 0 to 3, with a total score between 0 to 21. Participants with a score above 5 were considered poor sleepers.

The 4th part evaluated education-related activities; students were asked whether they had difficulty waking up, morning fatigue, poor grades, difficulty concentrating, class absence, and tardiness.

Statistical analysis

The statistical analysis was carried out using SPSS 23 (IBM Corp. Released 2015. IBM SPSS Statistics for Windows, Version 23.0. Armonk, NY: IBM Corp). Quantitative variables were reported as mean± standard deviation. Non-parametric data were represented in median and interquartile ranges, while qualitative variables were summarized in frequencies and percentages and compared utilizing the chi-square test. Logistic regression analysis was carried out to identify predictors for poor sleep quality among the

studied students. The level of significance was set at $p < 0.05$.

Ethical consideration

This study was approved by the ethical committee of the Faculty of Medicine for Girls, Al-Azhar University. Informed consent was obtained from each participant after clarification of the aim of the study and the confidentiality of the information.

Results

Students' profiles are depicted in Table 1 and Fig. 1. A total of 1184 medical students (Boys and Girls) at the Faculty of Medicine, Al Azhar University in Cairo, were recruited in the study. The mean age was 21.4±1.6, ranging between 18 to 24 years; 55.5% were boys, with the majority living in a hostel. About half reported consuming caffeinated beverages frequently, while only one-third were physically active. The BMI indicated that 48.8% were overweight or obese.

The smartphone use pattern is depicted in Table 2 and Fig. 2. All participants owned smartphones, the mean hours of mobile usage was 4.7± 0.5 ranging from 2 to 8 hours, and 96.5 % of students using smartphones at bedtime, with nearly half (46.9%) used them for more than 2 hours. Social media usage was the highest among students (74.2%), followed by communication (48.8%), and academic search usage was the lowest (30.3%). The majority of students do not adjust the screen's brightness or switch their phones off during sleep.

Table 3 and Fig. 3 depict the relationship between sleep patterns and smartphone use before bed. Approximately two-thirds (63.1%) of all participants had a PSQI score > 5 (they were considered poor sleepers), and 64.2% of bedtime smartphone users were poor sleepers. In addition, the median of PSQI was statistically significantly higher among bedtime smartphone users than among non-user students (7 versus 5, respectively) (p-value <0.05). By comparing phone bedtime users to non-users, it was found that about two-thirds (66.5%) faced difficulty falling asleep, 61% took more than thirty minutes, and 51.2% had interrupted sleep. In addition, 40.9% of them slept less than the recommended amount for their age (<7 h) and spent an average of 8.8±1.4 hours in bed, with statistically significant differences (p-value <0.05).

Regarding factors and predictors that were related to sleep quality, poor sleep was prevalent among male students (53.0%), those living in hostels (88.1%), those with a sedentary lifestyle (77.1%), students who regularly consumed caffeinated drinks after 4 p.m. (57.7%), overweight and obese students (52.7%), and students who did not turn off their phones during sleep (82.9%), with statistically significant differences compared to those who scored good sleep quality (p-value <0.05), as shown in Table 4.

Table 5 displays the results of a logistic regression analysis indicating that the likelihood of poor sleep quality was increased by the following significant predictors: bedtime smartphone use (OR=3.569), regular coffee consumption (OR=1.888), hostel residence (OR=1.1615), and phone proximity during sleep (OR=1.283). In contrast, physical activity was found to significantly reduce the likelihood of poor sleep quality (OR) by 0.605.

Sleep quality and academic activities are described in Table 6. Poor sleepers reported greater deterioration in academic achievement compared to good sleepers, as (82.7%) of poor sleepers faced difficulty in waking up, (79.3%) had morning fatigue, (65.3) attended classes late, (67.4) missed their classes, (82.0%) suffered from a decrease in concentration, and (79.7%) reported a decline in exam grades, with statistically significant differences (p-value <0.05).

Table 1 General profile of the studied students

Variables	Frequency (percentages) Total=1184
Age (yrs): (Mean ± SD)= 21.4 ±1.6	
≤ 21	555(46.9)
>21	629(53.1)
Sex	
Girls	527(44.5)
Boys	657(55.5)
Place of current residence	
With family	175(14.8)
At Hostel	1009(85.2)
Academic phase	
Pre clinical	413(34.9)
Clinical	771(65.1)
Practicing exercise	
Yes	321(27.1)
No	863(72.9)
Consumption of caffeinated drinks	
Rare	279(23.6)
Often	295(24.9)
Regular	610(51.5)
Current tobacco smoking	
Yes	38(3.2)
No	1146(96.8)
Body mass index	
Underweight	80(6.8)
Normal	526(44.4)
Overweight and obese	578(48.8)

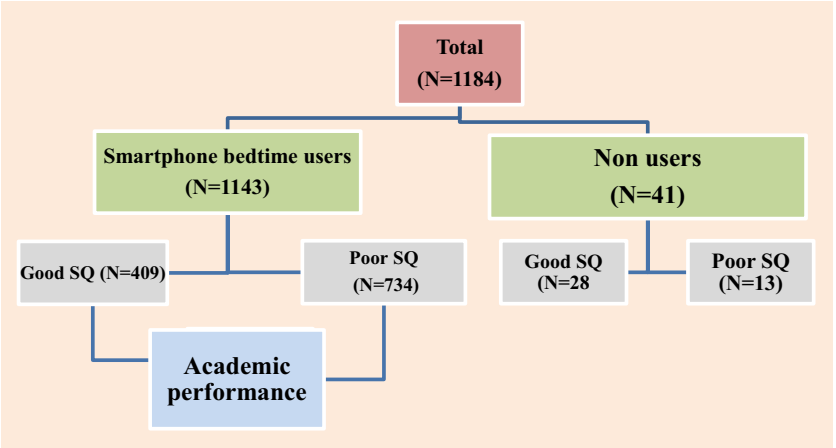


Fig. 1 Algorithm for the distribution of the studied students

Discussion

Smartphones enable us to utilize various applications that make our lives more organized. However, it can be somewhat distracting, particularly for adolescents and young adults, negatively affecting cognition and development. The present study focused on the smartphone usage patterns of medical students to determine the impact of smartphone use at bedtime on sleep quality and, consequently, academic performance.

The current study showed that all participants own smartphones with 4.7± 0.5 hours as mean daily usage ranged from 2 to 8 hours. This finding is in agreement with Rafique et al. (2020), who found that smartphone ownership was 98% among Sauidian University students. In addition, Meng et al. (2021) in China revealed that all medical participant students (recruited from all relevant medical majors and all years of medical school) had smartphones with nearly the same duration of daily mobile usage in China (5 hours) and longer duration (8.57 hours) in Saudi Arabia. The bedtime usage of smartphones was highly prevalent (96.5 %) among the study participants. This finding is compatible with Pham et al. (2021), Alshobaili and AlYousefi (2019), and Haque et al. (2017), who reported (98.1%, 92.6%, and 96.75%, respectively) bedtime usage among university students.

Smartphone overuse was defined as using more than one hour at bedtime (Boonluksiri 2018). About two-thirds of the participants were smartphone-over users (in which 48.6% of usage duration was >two hours, and 23.6% ranged from 1 to 2 hours). This result aligns with Meng et al. (2021), who found that 59% of college students used phones for more than an hour at bedtime.

Smartphones have enabled students to communicate with their colleagues and lecturers to share academic knowledge. It also enabled them to engage in social discussions and dialogue on different ideas and topics. They are also used for entertainment through gaming and watching videos. In the present study, social media was the most prevalent usage purpose (74.2%), followed by communication (48.8%), then watching videos (33.6%), while academic activity was presented by only 30.3%. In accordance, Haque et al. (2017) revealed that the preferred purposes of smartphone usage were social media apps (99.53%), educational apps (83.26%), video apps (74.42%), and gaming apps (31.63%). In addition, Gladius et al. (2018) found that 100% of their students used

Table 2 Pattern of phone use among smartphone bedtime-user students

Variables	Frequency (percentages) No.= 1143
Duration of daily use (hrs.) (Mean ± SD)	4.7±0.5
Duration of use at bedtime(hrs./day)	
<1	318(27.8)
1-2	270(23.6)
>2	555(48.6)
Purpose of use	
Social media	848(74.2)
Communication	558(48.8)
Watching videos	384(33.6)
Academic search	346(30.3)
Gaming	176(15.4)
Adjustment of screen display light at bedtime	
Yes	234(20.5)
No	909(79.5)
Switching off phone during sleep	
Yes	216(18.9)
No	927(81.1)
Position of phone during sleep	
At bed	670(58.6)
Near bed	473(41.4)

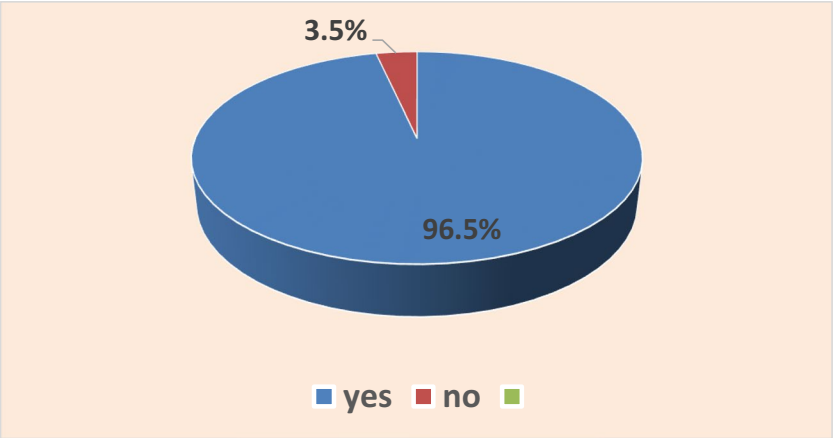


Fig. 2 Prevalence of smartphone use at bedtime among the studies students

mobiles as a mode of communication, followed by social media (66.5%) and only 31.5% for academic purposes. Moreover, Meng et al. (2021) mentioned that 91% of college students used mobile phones for entertainment and 61% to obtain information. After the dominance of online learning during the COVID pandemic, the traditional face-to-face learning method was reintroduced and resumed, which may account for the relatively low proportion of students who use smartphones for academic purposes.

Medical students are at higher risk of sleep disturbances and their consequences due to poor time management skills and high academic load (Copaja-Corzo et al. 2022). About two-thirds of students in the present study had poor sleep quality (63.1%). This finding was supported by a systematic review in which that poor sleep quality was reported among 65% of European medical students (Rao et al. 2020). Contradictory to the present study findings, Meng et al. (2021) and Tao et al. (2017) reported a lower prevalence (13% and 9.8%, respectively), which could be attributed to the variation in a medical and educational environment.

Many college students are unaware of the side effects of using smartphones at bedtime; they incorrectly imagine that these media might help them sleep, and their phone use becomes part of their sleep routine (Amra et al. 2017). The current study demonstrated that poor sleep quality was significantly higher among bedtime smartphone users (64.2%) than non-users (31.7%), and reported that smartphone users at bedtime had a significantly higher prevalence of increased sleep latency (time to fall asleep), difficulty in maintaining sleep (sleep

Table 3 Sleep pattern in relation to smartphone usage at bedtime among the studied students

	Bedtime smartphone use		Significance tests	*P*- value
	Non-users (41) N (%)	Users (1143) N (%)		
Pittsburgh sleep quality index (PSQI)				
Median (Q3-Q1)	5 (4-7)	7 (5-10)	Z=3.8	0.000*
Sleep latency				
15m	28(68.3)	178(15.6)	χ^2=82.5	0.000*
30m	10(24.4)	268(23.4)		
45m	1(2.4)	320(28.0)		
1 h	2(4.9)	377(33.0)		
Difficulty in falling asleep				
Yes	16(39.0)	760(66.5)	χ^2=13.2	0.000*
No	25(61.0)	383(33.5)		
Sleep disturbance				
Yes	13(31.7)	585(51.2)	χ^2=6	0.01*
No	28(68.3)	558(48.8)		
Actual sleep hours				
<7 h	7(17.1)	468(40.9)	χ^2=9.4	0.002*
7-9 h	34(82.9)	675(59.1)		
Hours spent in bed				
Mean ± SD	7.8±0.8	8.8±1.4	t=4.7	0.000*

*Statistically significant difference $p<0.05$

disturbances) and lower sleep duration in comparison to non-users (p-value<0.05). These findings are in agreement with other studies; Exelmans and Van den Bulck (2016) revealed that respondents who brought their mobile phones into the bedroom had significantly longer sleep latency, poorer sleep efficiency, more sleep disturbance, and greater daytime dysfunction.

Yang et al. (2019) reported that there was a gradual decline in sleep quality and duration, as well as an increase in sleep onset latency, due to the long-term effects of using light-emitting electronic devices in bed in his study that was conducted on students from six universities.

Also, Amra et al. (2017) found that adolescents who used cell phones after 9 p.m. had higher levels of sleep latency and delayed wake-up time than non-users. This finding is consistent with the current study's logistic regression model, which revealed that smartphone users before bedtime had a higher risk of poor sleep quality (OR= 3.6) than non-users. They were 1.39 times more likely to experience poor quality of sleep. In addition, Pham et al. (2021) demonstrated that the use of electronic devices within two hours of bedtime led to a reduction in sleep quality after controlling for confounding variables. In addition, Meng et al. 2021 found that the risk of poor sleep quality was 1.41–1.59 times greater for those who used their phones for more than 30 minutes before bed.

The previous result can be explained by many hypotheses; one of them suggested that the short-wavelength light (blue light) emitted by mobile phones inhibits melatonin production, thus stimulating cognitive functions and increasing brain alertness. Sleep displacement is another mechanism; with the convenience of smartphone use in bed, time passes without noticing and displaces sleep time (Alshobaili and AlYousefi 2019). A third proposed assumption suggested that emotional arousal, scanning for messages, text updates, seeing exciting something, or reading a negative notification can cause hyperexcitability and contribute to a hyperarousal state (Rafique et al. (2020). Therefore, the quality of sleep is significantly improved when the smartphone is turned off at bedtime.

Regarding demographic and lifestyle factors affecting sleep quality, the current study revealed that hostel-dwelling students had poorer sleep quality than others. Multiple factors, including peer pressure, additional stress, and homesickness, may account for the poorer sleeping habits of hostel residents. In accordance, Saat et al. (2020) discovered that the residential neighborhood is a risk factor for poor sleep quality.

Caffeine is the most widely used psychoactive substance for restoring wakefulness, modulating brain activity, and enhancing task performance (Khan et al., 2016). According to this study, frequent consumption of caffeinated beverages was associated with poor sleep quality. In agreement with this finding, Pham et al. (2021) and Al Shammari et al. (2020) suggested that consuming fewer caffeinated beverages earlier in the day is a more effective way to enhance sleep quality.

Moreover, this study demonstrated that those who did not exercise regularly had a higher rate of poor sleep quality than those who did this finding is consistent with Ozcan and Acimis (2021), who found that physical inactivity is an independent risk factor for poor sleep quality, aligning with Pham et al. (2021). Similarly, a systematic review done by Wang and Boros (2019) concluded that moderate exercise had a more favorable outcome on sleep quality than vigorous exercise through improved neuropsychological performance.

With regard to BMI, the current study found that poor sleep quality was significantly more prevalent among overweight and obese students than among normal-weight and underweight students. This result is consistent with the findings of Kristicevic et al. (2018), who demonstrated that poor sleep quality was more prevalent in cases with BMI ≥25 kg/m2 than in cases with BMI<25 kg/m2. This result can be attributed to the alterations in metabolism and/or sleep-wake cycles caused by obesity, which deteriorates sleep quality.

In contrast, in the current work it has been found that age, academic year, and smoking had no significant relationship with sleep quality. Abdallah et al. (2021) reported the same findings regarding age and smoking among Egyptian participants (also, no significant association between smoking and academic year with sleep quality were reported by Pham et al. 2021)).

The present work revealed insignificant differences between adjusting and not adjusting screen light mode on sleep quality, consistent with Pham et al. (2021). However, avoiding blue light exposure at bedtime is recommended through blocking glasses or filters to reduce sleep disturbances and improve sleep quality (Perez Algorta et al. 2018).

Good sleep is associated with good physical health, which is crucial for good academic performance. Poor sleep quality caused by excessive use of smartphones may influence students' concentration levels, decision-creating, memory and

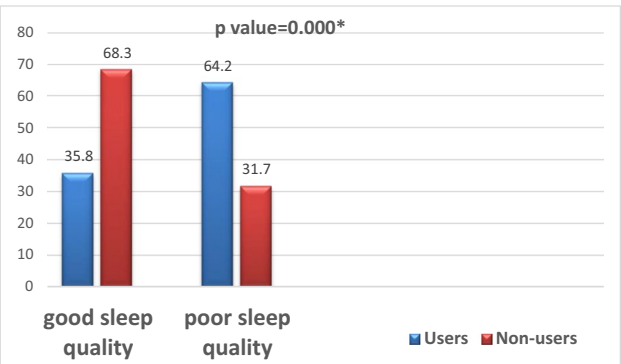


Fig. 3 Sleep quality in relation to bedtime smartphone use among the studied students

Table 4 Demographic and personal characteristics of studied students in relation to sleep quality score

Variables	Pittsburgh sleep quality index (PSQI)		Significance tests	*P*- value
	Good 437	Poor 747		
Age				
≤ 21	199(45.5)	356(47.7)	χ^2=0.5	0.48
>21	238(54.5)	391(52.3)		
Sex				
Girls	176(40.3)	351(47.0)	χ^2=5.03	0.02*
Boys	261(59.7)	396(53.0)		
Place of current residence				
Family	86(19.7)	89(11.9)	χ^2=13.2	0.000*
Hostel	351(80.3)	658(88.1)		
Academic phase				
Pre clinical	151(34.6)	262(35.1)	χ^2=0.03	0.85
Clinical	286(65.4)	485(64.9)		
Practicing exercise				
Yes	150(34.3)	171(22.9)	χ^2=18.2	0.000*
No	287(65.7)	576(77.1)		
Consumption of caffeinated drinks				
Rare	166(38.0)	113(15.1)	χ^2=80.3	0.000*
Often	92(21.1)	203(27.2)		
Regular	179(41.0)	431(57.7)		
Current tobacco smoking				
Yes	18(4.1)	20(2.7)	χ^2=1.8	0.17
No	419(95.9)	727(97.3)		
Body mass index				
Underweight	29(6.6)	51(6.8)	χ^2=13.7	0.001*
Normal	224(51.3)	302(40.4)		
Overweight and obese	184(42.1)	394(52.7)		
Adjust smartphone's screen display light				
Yes	101(23.1)	142(19.0)	χ^2=2.8	0.09
No	336(76.9)	605(81.0)		
Switching phone during sleep				
Yes	96(22.0)	128(17.1)	χ^2=4.2	0.04*
No	341(78.0)	619(82.9)		
Position of phone during sleep				
At bed	242(55.4)	452(60.5)	χ^2=2.9	0.08
Near bed	195(44.6)	295(39.5)		

*Statistically significant difference p<0.05

learning abilities, which may result in low grades. (Amra et al. 2017; Wagner et al. 2004; Maheshwari and Shaukat 2019).

With respect to the effect of sleep quality on academic activities, this study revealed that most poor sleeper students faced difficulty in waking up, had morning fatigue, suffered from a decrease in concentration, and reported a decline in exam grades. At the same time, about two-thirds of them attended classes late and missed their classes, with statistically significant differences compared to students demonstrating good sleep quality.

Table 5 Logistic regression of predictors of poor sleep quality among the studied students

Predictors	B	Sig.	OR	95% CI
Smartphone bedtime use	1.272	0.000*	3.569	1.798-7.082
Regular coffee drinking	0.636	0.000*	1.888	1.478-2.412
Physical activity	- 0.503	0.000*	0.605	0.462- 0.790
Hostel residence	0.480	0.005*	1.615	1.156 -2.258
Position of phone near bed during sleep	0.249	0.048*	1.283	1.002-1.643

*Statistically significant difference p<0.05

Table 6 Academic performance in relation to sleep quality among bedtime smartphone users

Variables	PSQI		Significance tests	*P*- value
	Good 409	Poor 734		
Difficulty in waking up				
Yes	313(76.5)	607(82.7)	χ^2=6.4	0.01*
No	96(23.5)	127(17.3)		
Morning fatigue				
Yes	244(59.7)	582(79.3)	χ^2=50.5	0.000*
No	165(40.3)	152(20.7)		
Being late for classes				
Yes	199(48.7)	479(65.3)	χ^2=30	0.000*
No	210(51.3)	255(34.7)		
Increased missed classes				
Yes	192(46.9)	495(67.4)	χ^2=46	0.000*
No	217(53.1)	239(32.6)		
Decrease in concentration at daytime				
Yes	283(69.2)	602(82.0)	χ^2=24.7	0.000*
No	126(30.8)	132(18.0)		
Decline in exam grades				
Yes	235(57.5)	585(79.7)	χ^2=64.1	0.000*
No	174(42.5)	149(20.3)		

*Statistically significant difference p<0.05

In line, Gladius et al. (2018) found a significant association between poor sleep quality with mobile phone usage and academic performance (p<0.01). Similarly, Rathakrishnan et al. (2021) showed a negative relation between sleep quality and academic performance among Malaysian students. Furthermore, Elbilgahy et al. (2021) found a significant association between sleep disorders and academic performance among Egyptian and Saudi students related to internet addiction.

Strengths and limitations

The current study has many strengths, a large sample size that includes students from both male and female medical schools and all academic levels. In addition, the participants were selected at random, allowing the findings to be generalized to the remaining students. In addition, the prevalence of smartphone use before bed and poor sleep quality were demonstrated by our research. We further investigated the relationship between smartphone use before bed, sleep quality, and educational activities. Nevertheless, certain limitations should be reported. First, due to the cross-sectional design, it was not possible to examine casual associations between smartphone use before bed, poor sleep quality, and academic disruption. Second, sleep quality and duration of smartphone use were self-reported by participants, which may have led to recording bias.

Conclusion

This study revealed that smartphone use at bedtime was highly prevalent among the medical students surveyed. In addition, poor sleep quality was prevalent among study participants, and it was significantly higher among bedtime smartphone users than among non-users, making this a significant health issue that negatively impacts academic activities. Therefore, it is crucial to educate medical students about the negative effects of smartphone use before bedtime on sleep and the importance of adequate sleep for good academic performance.

Acknowledgments The authors acknowledged all students who shared in the study.

Author contribution All authors fulfill author contribution criteria

Funding Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB).

Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Code availability (software application or custom code) Not applicable' for that section.

Declarations

Ethical approval The study was approved from the ethical committee of the Faculty of Medicine for girls, Al-Azhar University. Informed consent was obtained from each participant after clarification of the aim of the study and confidentiality of the information.

Consent to participate Informed consent was obtained from each participant in the study

Consent for publication Not applicable' for that section

Conflict of interest The authors declare no conflicts of interest.

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